🏷️ ArUco / AprilTag Visual Spatial Relocalization Navigation

1. Course Content

- Master using AprilTag / ArUco spatial visual anchors combined with camera PnP pose solving to achieve millimeter-level global rapid relocalization and navigation for mobile robots.
- Master the closed-loop workflow of automatically triggering pose recalibration when laser SLAM localization drift or "robot kidnapping" occurs by recognizing tag codes in the field of view.
- Master the recording calibration (Record mode) of tag codes in the global map coordinate system and the operation of real-time self-healing relocalization (Relocalize mode).
- **This case uses 3 * 3 * 3 AprilTag wooden blocks**

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2. Visual Spatial Relocalization Principle Analysis

In actual industrial AGV/AMR scenarios, lidar localization is prone to feature degradation causing localization step loss or jumping in long straight corridors, large open areas, or dynamic crowd interference; additionally, when the robot is manually transported (robot kidnapping), traditional algorithms have difficulty instantaneously recovering the true pose.

This system uses high-precision AprilTag / ArUco visual tag codes arranged in the environment as spatial global absolute anchors:

1. **PnP Spatial Pose Solving**: Onboard camera real-time detects tag corners in the image, combined with camera intrinsic matrix, solves the six-degree-of-freedom spatial transformation of the tag relative to the camera through Perspective-n-Point (PnP) algorithm.
2. **Spatial Coordinate System Inverse Chain Solving**:
   Combined with the tag's global pose pre-recorded in the map, the robot's absolute pose in the map is inversely calculated through homogeneous matrix transformation.

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3. Preparation and Running (Phase 1: Record Tag Code Global Position)

Before formally using visual relocalization:

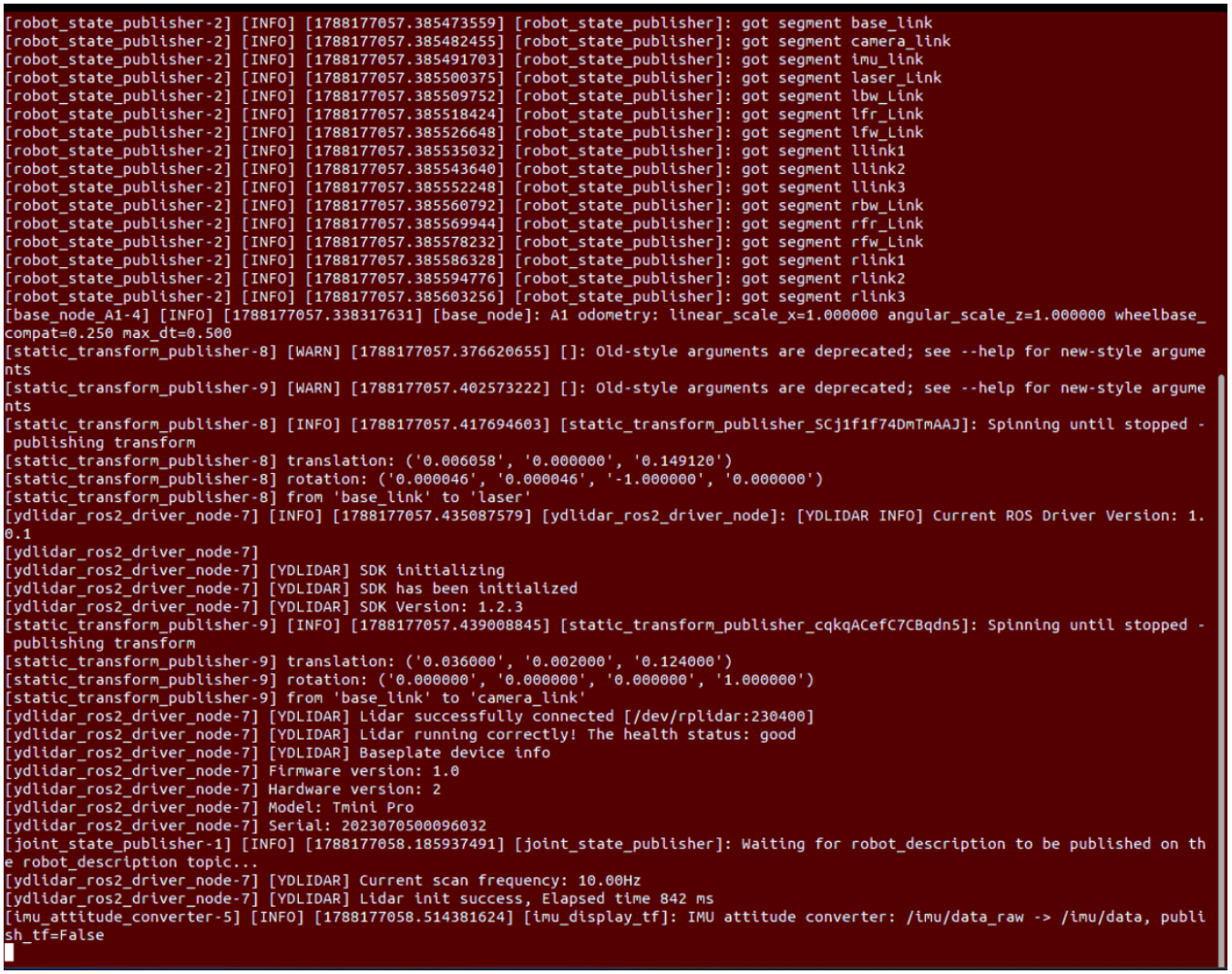
- It is necessary to first record and save the global coordinates of the tag codes arranged on walls or floors into the configuration file in a high-precision lidar localization environment.
- It is necessary to first learn Chapter 7 (Cartographer Algorithm Mapping), build the yaml map and pose map.

This phase requires starting the following nodes in sequence:

3.1 Start Chassis and Sensors

Run in **Terminal 1** (start chassis driver, lidar, and RGB depth camera):

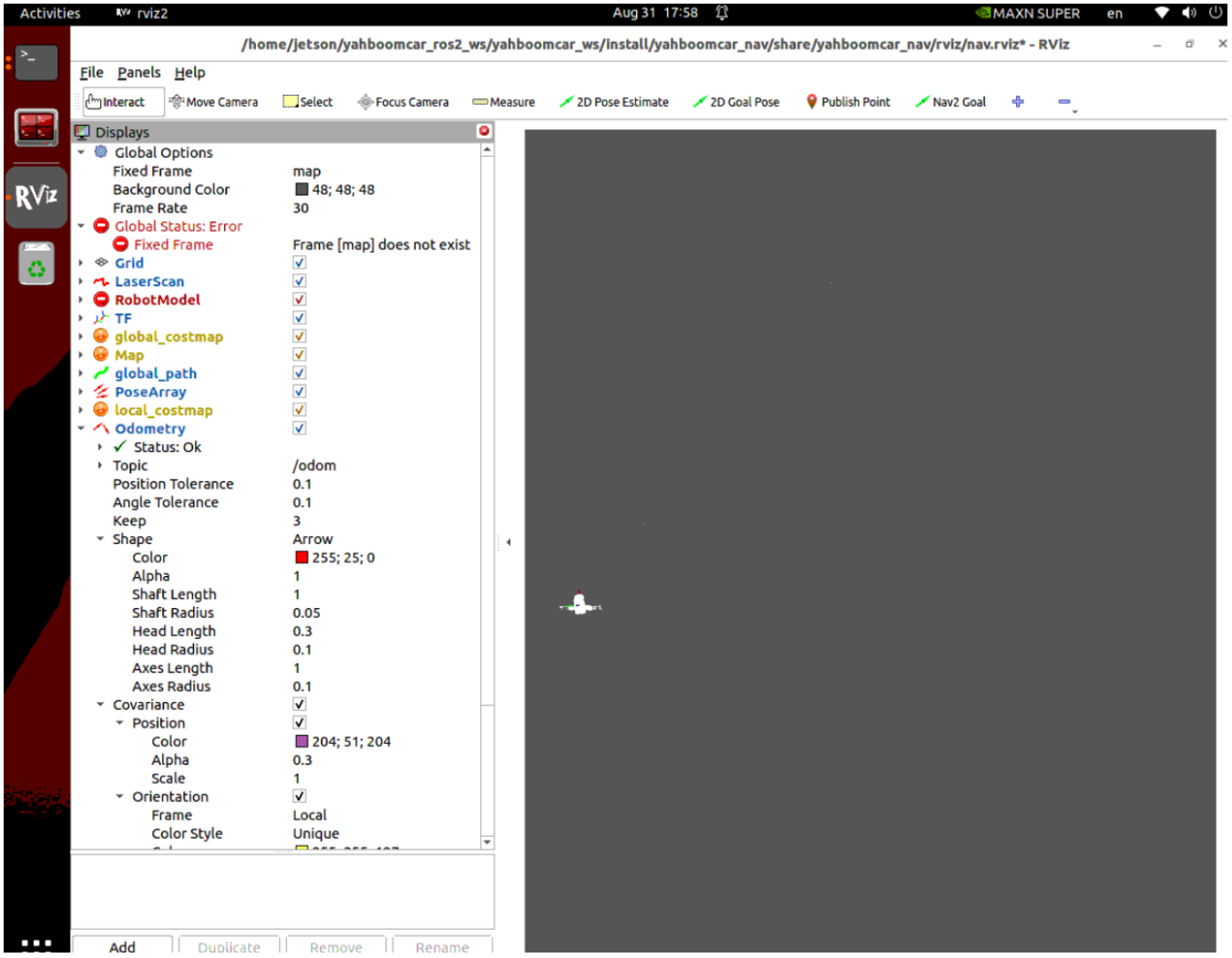
```
ros2 launch yahboomcar_nav laser_bringup_launch.py
```



3.2 Start RViz Navigation Visualization

Run in **Terminal 2** (it is recommended to open RViz on the virtual machine if the VM and robot are on the same LAN):

```
ros2 launch yahboomcar_nav display_nav_launch.py
```



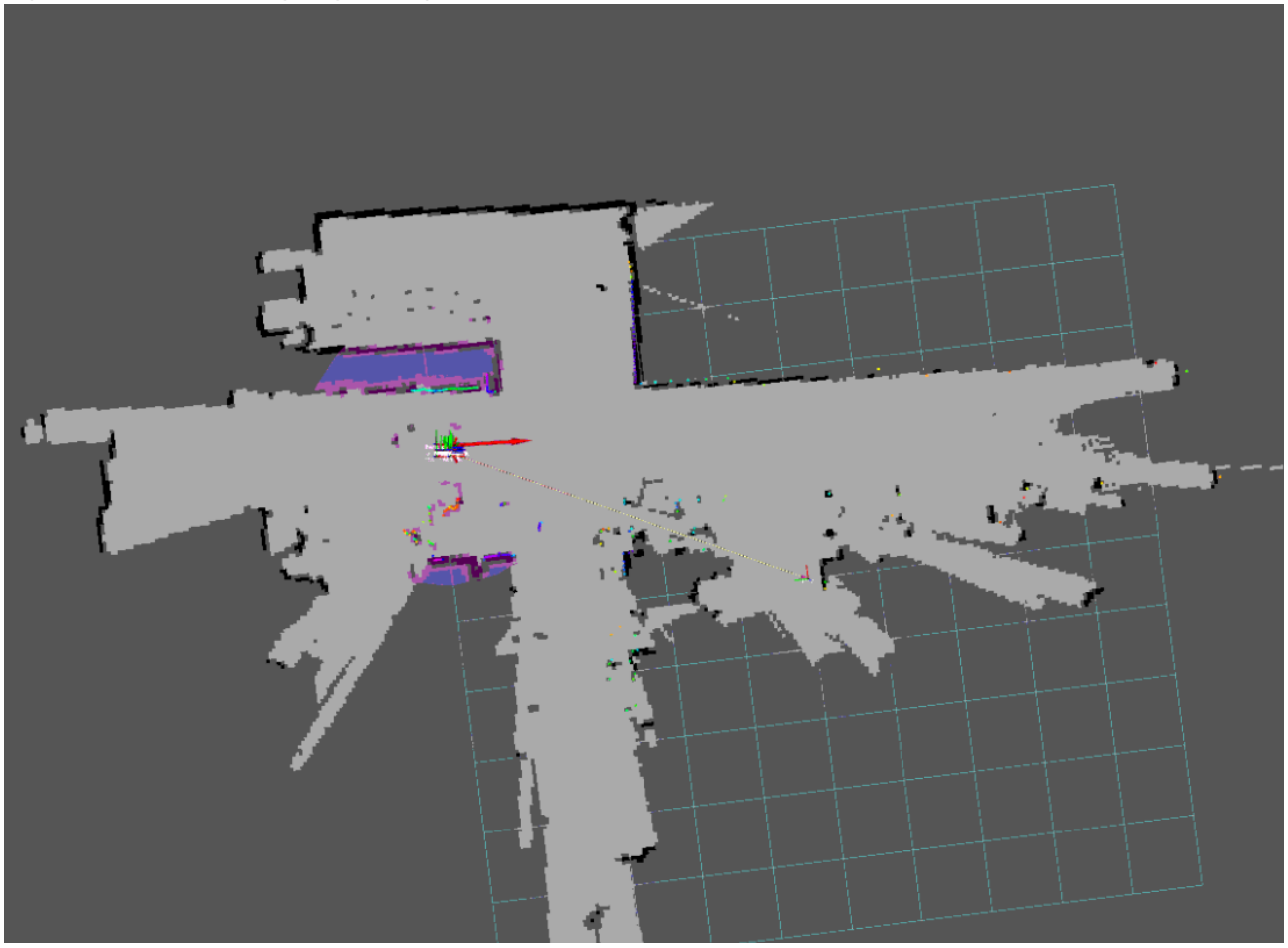
3.3 Start Cartographer Pure Localization Node

Run in **Terminal 3** to load `.pbstream` pose graph and start pure localization core:

```
ros2 launch yahboomcar_nav localization_imu_odom.launch.py use_rviz:=false
load_state_filename:=$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/map.pbst
ream
```

```
jetson@yahboom:~$ ros2 launch yahboomcar_nav localization_imu_odom.launch.py use_rviz:=false load_state_filename:=/home/jetson/map/map.pbstream
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/pose_graph.luapose_graph.lua'.
[cartographer_node-1] [INFO] [1788170604.259213680] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/pose_graph.luapose_graph.lua'.
[cartographer_node-1] [INFO] [1788170604.259476990] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder.luatrajectory_builder.lua'.
[cartographer_node-1] [INFO] [1788170604.259519520] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder.luatrajectory_builder.lua'.
[cartographer_node-1] [INFO] [1788170604.259595909] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_2d.luatrajectory_builder_2d.lua'.
[cartographer_node-1] [INFO] [1788170604.259629734] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_2d.luatrajectory_builder_2d.lua'.
[cartographer_node-1] [INFO] [1788170604.259807408] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_3d.luatrajectory_builder_3d.lua'.
[cartographer_node-1] [INFO] [1788170604.259843922] [cartographer logger]: I0831 18:03:24.000000 70766 configuration_file_resolver.cc:126] Found '/home/jetson/yahbo
omcar_ros2_ws/software/library_ws/install/cartographer/share/cartographer/configuration_files/trajectory_builder_3d.luatrajectory_builder_3d.lua'.
[cartographer_node-1] [INFO] [1788170604.311704267] [cartographer logger]: I0831 18:03:24.000000 70766 map_builder_bridge.cpp:117] Loading saved state '/home/jetson
/map/map.pbstream'...
[cartographer_node-1] [INFO] [1788170604.518953543] [cartographer logger]: I0831 18:03:24.000000 70766 map_builder_bridge.cpp:136] Added trajectory with ID '1'.
[cartographer_node-1] [INFO] [1788170604.565510241] [cartographer logger]: I0831 18:03:24.000000 70766 ordered_multi_queue.cc:172] All sensor data for trajectory 1
is available starting at '639237674045638968'.
[cartographer_node-1] [INFO] [1788170604.594751697] [cartographer logger]: I0831 18:03:24.000000 70766 local_trajectory_builder_2d.cc:312] Extrapolator not yet init
ialized.
[cartographer_node-1] [INFO] [1788170604.645255745] [cartographer logger]: I0831 18:03:24.000000 70766 local_trajectory_builder_2d.cc:124] Extrapolator not yet init
ialized.
[cartographer_node-1] [INFO] [1788170604.707386407] [cartographer logger]: I0831 18:03:24.000000 70766 local_trajectory_builder_2d.cc:135] Extrapolator is still ini
tializing.
[cartographer_node-1] [INFO] [1788170604.795973862] [cartographer logger]: I0831 18:03:24.000000 70766 pose_graph_2d.cc:148] Inserted submap (1, 0).
```

> **Tip**
>
> After pure localization starts, if in a weak feature area, you can use the `2D Pose Estimate` tool in RViz top toolbar to assist aligning initial pose:

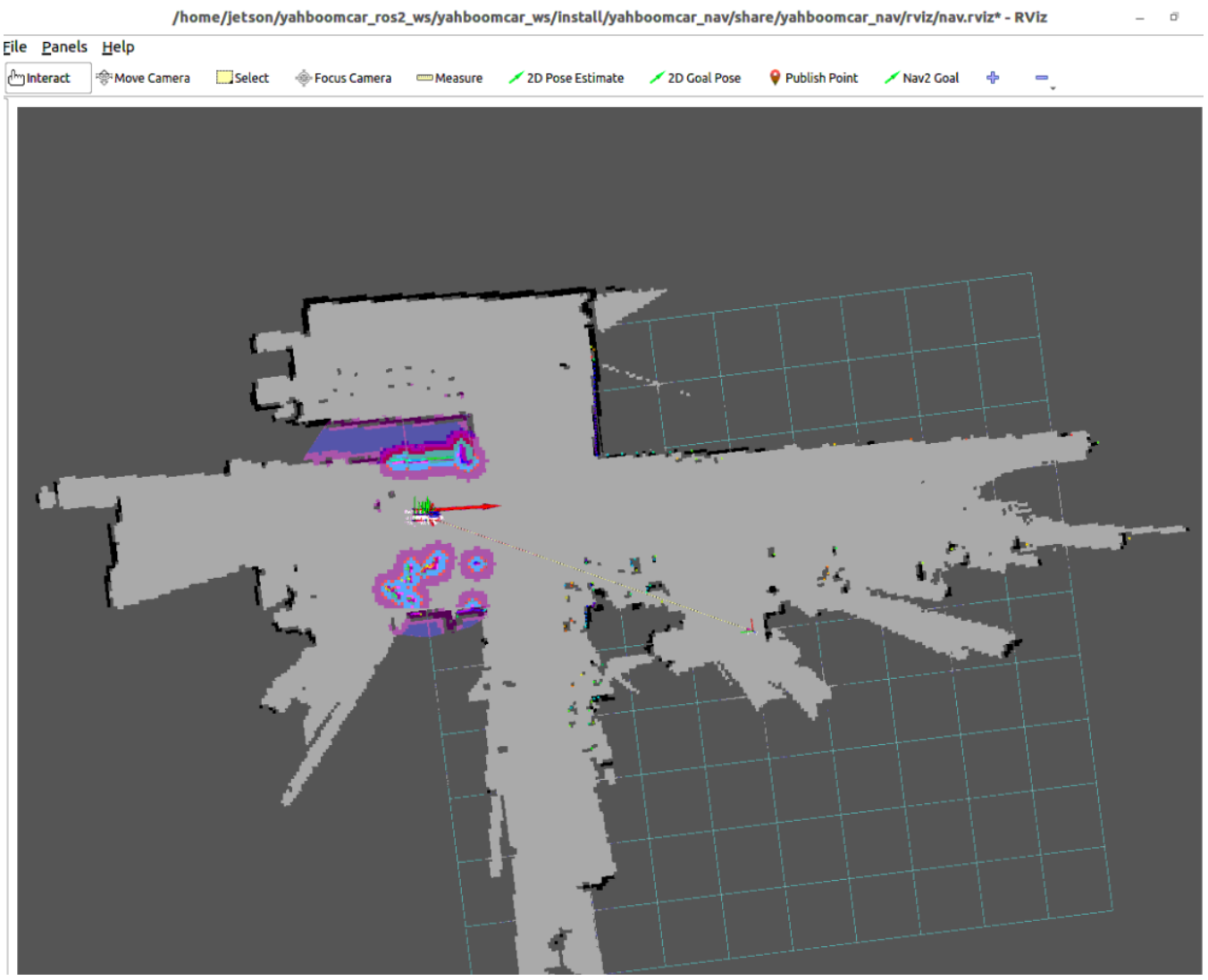


3.4 Start Nav2 Navigation Controller

Run in **Terminal 4** to start navigation controller (loads corresponding static raster map):

```
ros2 launch yahboomcar_nav navigation_cartodwb_launch.py
maps:=$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/maps/map.yaml
```

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws$ ros2 launch yahboomcar_nav navigation_cartodwb_launch.py maps:=/home/jetson/map/map.yaml
[component_container_isolated-1] [INFO] [1788170672.360287403] [behavior_server]: Creating bond (behavior_server) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.467487545] [lifecycle_manager_navigation]: Server behavior_server connected with bond.
[component_container_isolated-1] [INFO] [1788170672.467628992] [lifecycle_manager_navigation]: Activating bt_navigator
[component_container_isolated-1] [INFO] [1788170672.469218311] [bt_navigator]: Activating
[component_container_isolated-1] [INFO] [1788170672.608745067] [bt_navigator]: Creating bond (bt_navigator) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.739241174] [lifecycle_manager_navigation]: Server bt_navigator connected with bond.
[component_container_isolated-1] [INFO] [1788170672.739378300] [lifecycle_manager_navigation]: Activating waypoint_follower
[component_container_isolated-1] [INFO] [1788170672.740677206] [waypoint_follower]: Activating
[component_container_isolated-1] [INFO] [1788170672.740783163] [waypoint_follower]: Creating bond (waypoint_follower) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.847842019] [lifecycle_manager_navigation]: Server waypoint_follower connected with bond.
[component_container_isolated-1] [INFO] [1788170672.847954760] [lifecycle_manager_navigation]: Activating velocity_smoother
[component_container_isolated-1] [INFO] [1788170672.848561411] [velocity_smoother]: Activating
[component_container_isolated-1] [INFO] [1788170672.848706122] [velocity_smoother]: Creating bond (velocity_smoother) to lifecycle manager.
[component_container_isolated-1] [INFO] [1788170672.952698535] [lifecycle_manager_navigation]: Server velocity_smoother connected with bond.
[component_container_isolated-1] [INFO] [1788170672.952832077] [lifecycle_manager_navigation]: Managed nodes are active
[component_container_isolated-1] [INFO] [1788170672.952858926] [lifecycle_manager_navigation]: Creating bond timer...
[component_container_isolated-1] [INFO] [1788170674.245692524] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 281 X 376 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170683.245718747] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 281 X 376 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170688.245735423] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170691.245707579] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170699.245692377] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170708.245719346] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170716.245724174] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170724.245696744] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170732.245724930] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170741.245882878] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 283 X 380 at 0.050000 m/pix
[component_container_isolated-1] [INFO] [1788170749.245696157] [global_costmap.global_costmap]: StaticLayer: Resizing costmap to 278 X 380 at 0.050000 m/pix
```



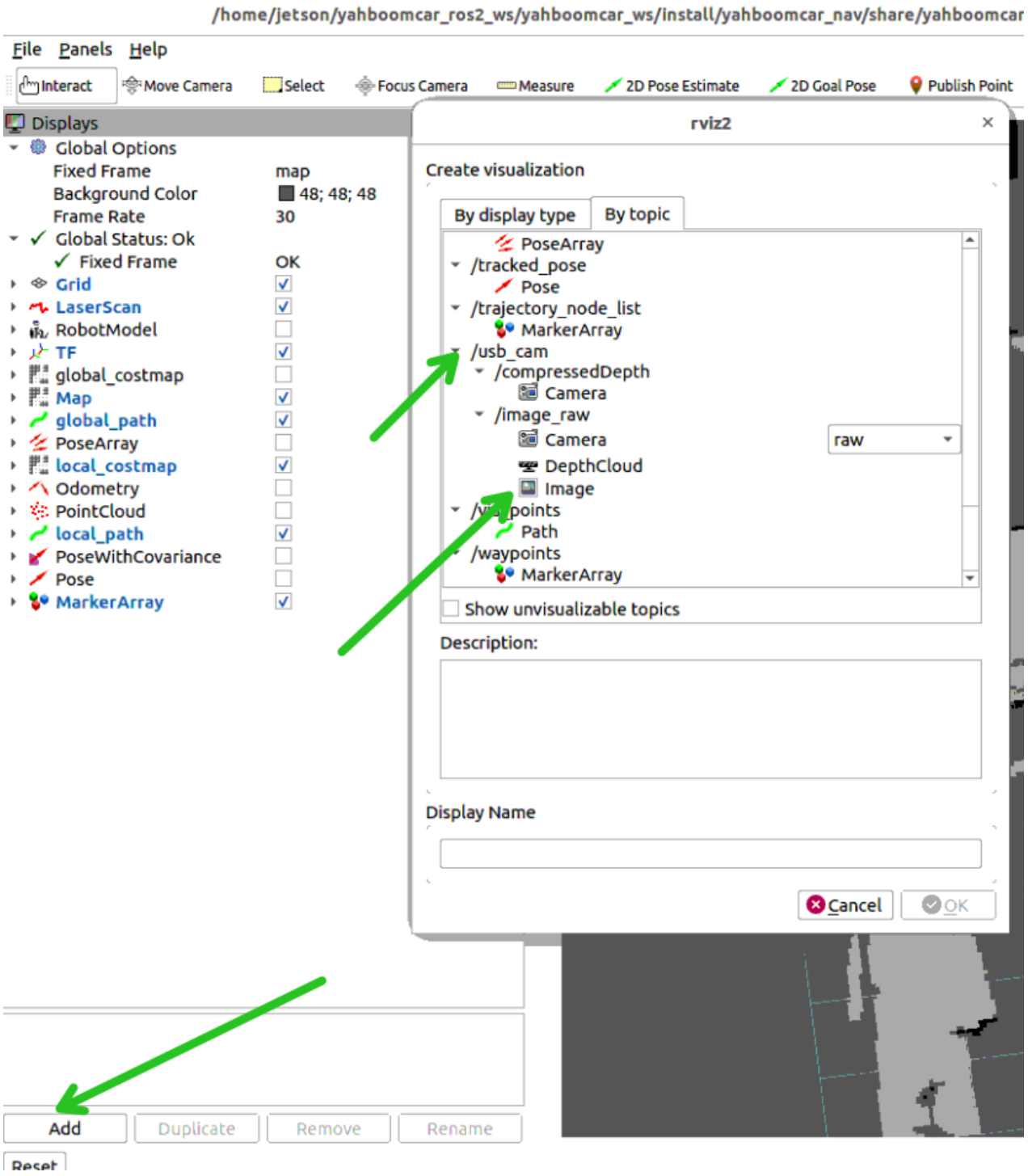
3.5 Start AprilTag Recording Mode

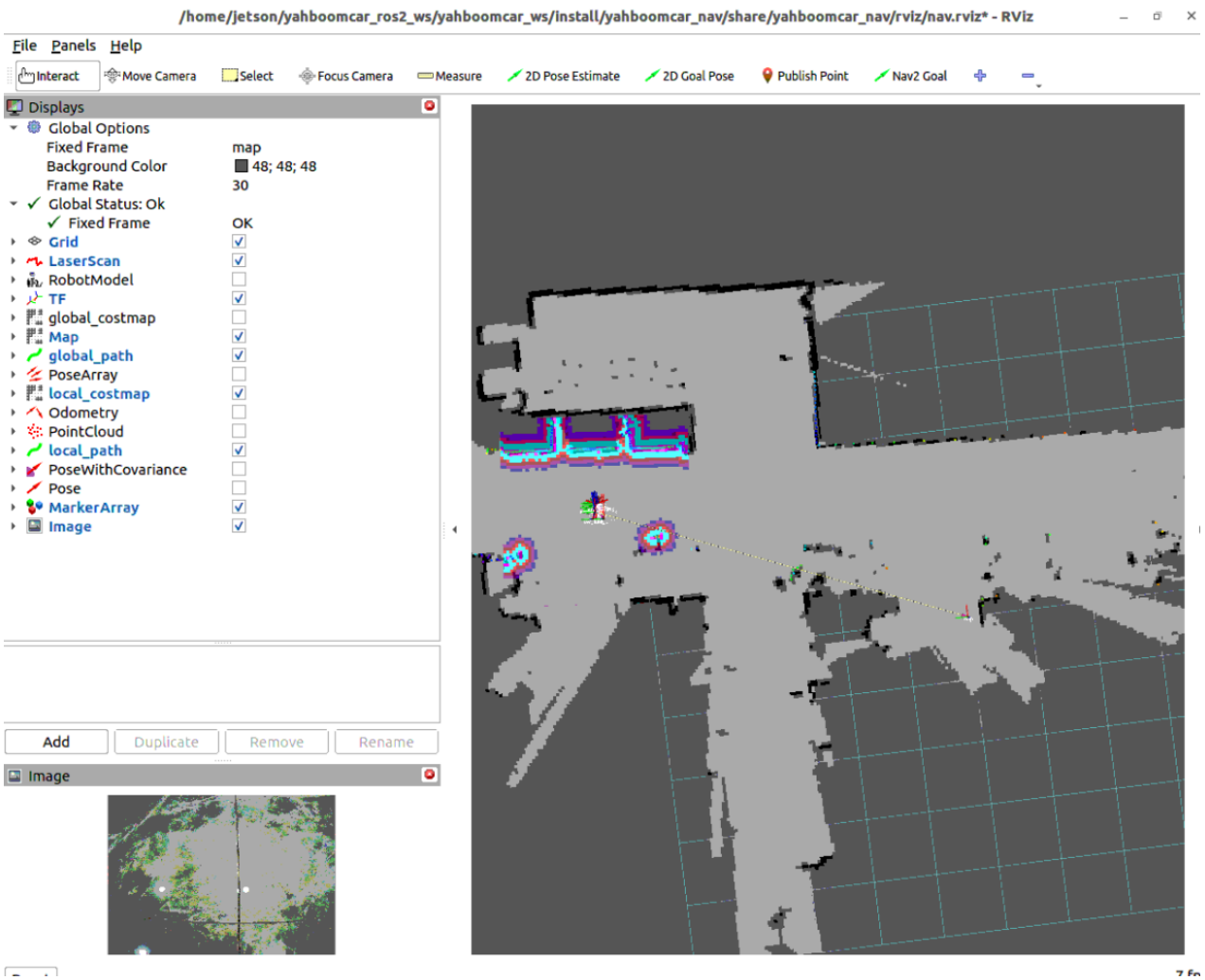
Run in **Terminal 5** (start detection node and pose recorder):

```
ros2 launch yahboomcar_nav apriltag_relocation.launch.py mode:=record
```

```
jetson@yahboom:~/yahboomcar_ros2_ws/yahboomcar_ws$ ros2 launch yahboomcar_nav apriltag_relocation.launch.py mode:=record
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2026-08-31-21-00-51-195326-yahboom-225659
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [apriltag_node-1]: process started with pid [225734]
[INFO] [apriltag_recorder-2]: process started with pid [225736]
[apriltag_recorder-2] [INFO] [1788181251.453186133] [apriltag_recorder]: AprilTag Recorder initialized
[apriltag_recorder-2] [INFO] [1788181251.453375100] [apriltag_recorder]: Output file: /home/jetson/yahboomcar_ros2_ws/dofbot_pro_ws/insta
l/dofbot_pro_bringup/share/dofbot_pro_bringup/config/aritag_relocation.yaml
[apriltag_recorder-2] [INFO] [1788181251.453399292] [apriltag_recorder]: Target frame: map
[apriltag_recorder-2] [INFO] [1788181251.459118402] [apriltag_recorder]: Service 'save_april_tag_pose' is ready
[apriltag_node-1] [INFO] [1788181251.839750346] [apriltag]: AprilTagNode initialized. Detection enabled: true
```

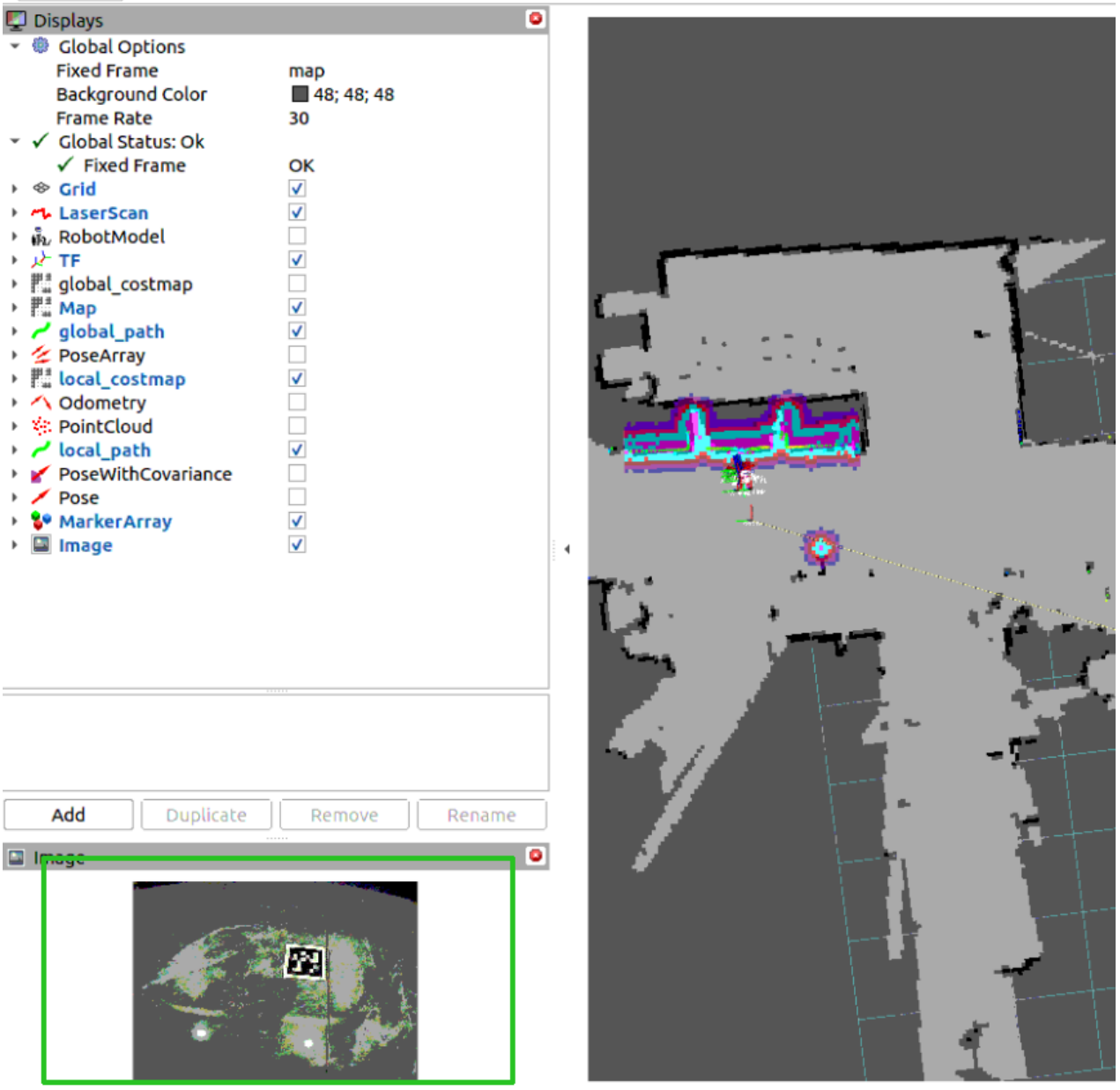
1. In RViz2 interface, click `Add`, select `By topic` → `/apriltag/image_rect` to add Image plugin to view camera detection feed in real-time:


/home/jetson/yahboomcar_ros2_ws/yahboomcar_ws/install/yahboomcar_nav/share/yahboomcar
File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Displays
Global Options
Fixed Frame map
Background Color 48; 48; 48
Frame Rate 30
Global Status: Ok
Fixed Frame OK
Grid
LaserScan
RobotModel
TF
global_costmap
Map
global_path
PoseArray
local_costmap
Odometry
PointCloud
local_path
PoseWithCovariance
Pose
MarkerArray
rviz2
Create visualization
By display type
By topic
PoseArray
/tracked_pose
Pose
/trajectory_node_list
MarkerArray
/usb_cam
/compressedDepth
Camera
/image_raw
Camera
raw
DepthCloud
Image
Path
/waypoints
MarkerArray
Show unvisualizable topics
Description:
Display Name
Cancel
OK
Add
Duplicate
Remove
Rename
Reset



2. Remote control the robot to drive to the front of the tag code; ensure the camera image clearly

observes the tag code and shows 3D coordinate axes and green recognition box:



3.6 View Tag ID and Record Pose

Run in **Terminal 6** to view the currently detected tag ID:

```
ros2 topic echo /detections --once
```

```
jetson@yahboom:~$ ros2 topic echo /detections --once
header:
  stamp:
    sec: 1788228493
    nanosec: 76618000
  frame_id: camera
detections:
- family: tag36h11
  id: 9
  hamming: 0
  goodness: 0.0
  decision_margin: 168.85043334960938
  centre:
    x: 359.7901492298834
    y: 297.025450250849
  corners:
  - x: 315.84664916992193
    y: 336.3739929199219
  - x: 402.0561218261719
    y: 338.699951171875
  - x: 400.7146911621093
    y: 260.3801879882812
  - x: 319.6192016601563
    y: 257.41665649414057
  homography:
  - 43.613582999050436
  - -11.565333670585568
  - 359.7901492298834
  - 2.841985557526524
  - 30.224905171716784
  - 297.025450250849
  - 0.005079352268773405
  - -0.030493026413103946
  - 1.0
---
jetson@yahboom:~$
```

After confirming the detected ID (for example ID is 9), call the service to persistently write the tag's global coordinates in the map into the configuration file:

```
ros2 service call /save_april_tag_pose apriltag_localization/srv/SaveAprilTagPose "{tag_id: 9}"
```

```
jetson@yahboom:~$ ros2 service call /save_april_tag_pose apriltag_localization/srv/SaveAprilTagPose "{tag_id: 9}"
requester: making request: apriltag_localization.srv.SaveAprilTagPose_Request(tag_id=9)

response:
apriltag_localization.srv.SaveAprilTagPose_Response(success=True, message='Successfully saved AprilTag ID 9', pose=geometry_msgs.msg.PoseStamped(header=std_msgs.msg.Header(stamp=builtin_interfaces.msg.Time(sec=1788228530, nanosec=424794342), frame_id='map'), pose=geometry_msgs.msg.Pose(position=geometry_msgs.msg.Point(x=3.5236221556951195, y=4.755936862879148, z=0.4118980485617142), orientation=geometry_msgs.msg.Quaternion(x=0.9505827987957547, y=-0.0271954387960573, z=-0.016506792884015867, w=0.3088369740348518))))

jetson@yahboom:~$
```

> **Note**
>
> After successful recording, the system will write that tag's precise pose in `map` coordinate system into `aritag_relocation.yaml`. If there are multiple tag codes, sequentially remote control the robot to the front of each tag and call the above service.

---

4. Phase 2: Visual Relocalization Autonomous Navigation Practical Test

After completing recording of all tag codes, close Terminal 5's recording program, and then enter the visual-assisted relocalization navigation phase.

4.1 Start Visual Relocalization Mode

Run in **Terminal 5**:

```
ros2 launch yahboomcar_nav apriltag_relocation.launch.py mode:=relocalize
```

```
jetson@yahboom:~$ ros2 launch yahboomcar_nav apriltag_relocation.launch.py mode:=relocalize
[INFO] [launch]: All log files can be found below /home/jetson/.ros/log/2026-09-01-10-09-22-323028-yahboom
-142414
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [apriltag_node-1]: process started with pid [142484]
[INFO] [apriltag_relocalization-2]: process started with pid [142486]
[apriltag_relocalization-2] [INFO] [1788228562.680206777] [apriltag_relocalization]: AprilTag Relocalizati
on node starting...
[apriltag_relocalization-2] [INFO] [1788228562.683028919] [apriltag_relocalization]:   aritag record file
  : /home/jetson/yahboomcar_ros2_ws/dofbot_pro_ws/install/dofbot_pro_bringup/share/dofbot_pro_bringup/conf
ig/aritag_relocation.yaml
[apriltag_relocalization-2] [INFO] [1788228562.699018609] [apriltag_relocalization]:   map_frame  : map
[apriltag_relocalization-2] [INFO] [1788228562.700753687] [apriltag_relocalization]:   base_frame : base_f
ootprint
[apriltag_relocalization-2] [INFO] [1788228562.702177334] [apriltag_relocalization]:   camera_frame: camer
a
[apriltag_relocalization-2] [INFO] [1788228562.703499826] [apriltag_relocalization]:   cooldown   : 5.0 s
[apriltag_relocalization-2] [INFO] [1788228562.709724473] [apriltag_relocalization]:   Loaded tag 11: pos=
[3.722, -0.154, 0.025]
[apriltag_relocalization-2] [INFO] [1788228562.711432478] [apriltag_relocalization]:   Loaded tag 12: pos=
[-0.952, 0.092, 0.025]
[apriltag_relocalization-2] [INFO] [1788228562.712384275] [apriltag_relocalization]:   Loaded tag 19: pos=
[-0.566, -0.024, 0.047]
[apriltag_relocalization-2] [INFO] [1788228562.713909396] [apriltag_relocalization]:   Loaded tag 9: pos=[
3.524, 4.756, 0.412]
[apriltag_relocalization-2] [INFO] [1788228562.733451868] [apriltag_relocalization]: Ready. Loaded 4 tag(s
) from map.
[apriltag_node-1] [INFO] [1788228563.034805538] [apriltag]: AprilTagNode initialized. Detection enabled: t
rue
[apriltag_relocalization-2] [INFO] [1788228563.080564578] [apriltag_relocalization]: Detected map tag 9 —
attempting relocalization...
[apriltag_relocalization-2] [INFO] [1788228563.091297259] [apriltag_relocalization]: Tag 9: published /ini
tialpose  pos=[3.791, 4.700, 0.318]  (dist=0.27m)
[apriltag_relocalization-2] [INFO] [1788228568.090749622] [apriltag_relocalization]: Detected map tag 9 —
attempting relocalization...
[apriltag_relocalization-2] [INFO] [1788228568.091089693] [apriltag_relocalization]: Tag 9: published /ini
```

The terminal will print relocalization logs:

```
[apriltag_relocalization-2] [INFO] [1788228563.080564578] [apriltag_relocalization]:
Detected map tag 9 — attempting relocalization...
[apriltag_relocalization-2] [INFO] [1788228563.091297259] [apriltag_relocalization]: Tag 9:
published /initialpose  pos=[3.791, 4.700, 0.318]  (dist=0.27m)
```

4.2 Visual Self-Healing Relocalization and Autonomous Navigation

1. **Simulate Localization Deviation/Kidnapping**:
   In RViz2, intentionally use `2D Pose Estimate` to set the robot's initial pose to a clearly erroneous deviating position; at this time, the lidar point cloud and map contours have serious misalignment.
2. **Instant Self-Healing Alignment**:
   Remote control the robot to rotate or move; as long as any recorded AprilTag tag code enters the camera field of view, the `apriltag_relocalization` node will, within milliseconds, calculate the true pose through TF chain solving and publish to `/initialpose`, and the robot model and lidar point cloud instantly restore alignment.
3. **Goal Point Autonomous Obstacle-Avoiding Navigation**:
   Click the `Nav2 Goal` (or `2D Goal Pose`) tool at the top of RViz to give a goal point on the map; the robot, based on the precisely calibrated pose, smoothly drives to the destination while avoiding obstacles.

> **Note**
>
> When the experiment ends, press `Ctrl+C` in each terminal window to stop the program.

---

5. Core Source Code and Algorithm Parameter Analysis

5.1 Visual Relocalization Unified Scheduling Launch Analysis

File Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/launch/nav2/apriltag_relocation.launch.py
```

Launch Core Design Analysis

```python
def generate_launch_description():
    ...
    # 1. Automatically adapt camera type (Nuwa / USB)
    # Dynamically select image topic and optical coordinate system based on camera_type
    # Nuwa camera: /ascamera_hp60c/camera_publisher/rgb0/image, camera_color_optical_frame
    # USB camera:  /usb_cam/image_raw, camera

    # 2. AprilTag core detection node (both modes start)
    apriltag_detect = Node(
        package='apriltag_localization',
        executable='apriltag_node',
        name='apriltag',
```

```
        parameters=[param_file, {'detection_enabled':
LaunchConfiguration('detection_enabled')}],
        remappings=[
            ('image_rect', LaunchConfiguration('image_topic')),
            ('image_rect/camera_info', LaunchConfiguration('camera_info_topic')),
        ],
        output='screen',
    )

    # 3. record mode: Start pose recorder node
    apriltag_recorder = Node(
        package='apriltag_localization',
        executable='apriltag_recorder',
        name='apriltag_recorder',
        output='screen',
        parameters=[{
            'output_file_path': LaunchConfiguration('aritag_record_file'),
            'target_frame': LaunchConfiguration('target_frame'),
            'camera_frame': LaunchConfiguration('camera_frame'),
        }],
        condition=IfCondition(is_record),
    )

    # 4. relocalize mode: Start relocalization solving and pose publishing node
    relocalization_node = Node(
        package='apriltag_localization',
        executable='apriltag_relocalization',
        name='apriltag_relocalization',
        output='screen',
        parameters=[{
            'aritag_record_file': LaunchConfiguration('aritag_record_file'),
            'cooldown_duration': LaunchConfiguration('cooldown_duration'),
            'map_frame': LaunchConfiguration('map_frame'),
            'robot_base_frame': LaunchConfiguration('robot_base_frame'),
            'camera_frame': LaunchConfiguration('camera_frame'),
        }],
        condition=IfCondition(is_relocalize),
    )
```

5.2 Spatial Relocalization Solving Core Source Analysis

File Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/apriltag_localization/src/apriltag_relocalization
.cpp
```

Spatial Homogeneous Transformation and Pose Publishing Core Implementation

```
bool AprilTagRelocalization::doRelocalize(int tag_id, const TagPose & map_pose, const
rclcpp::Time & stamp)
{
    std::string tag_frame = "tag" + std::to_string(tag_id);

    // 1. Get camera-to-tag real-time TF: T_camera_tag (output by AprilTag detection node
PnP solving)
    geometry_msgs::msg::TransformStamped tf_cam_tag = tf_buffer_.lookupTransform(
        camera_frame_, tag_frame, tf2::TimePointZero, tf2::durationFromSec(0.5));

    // 2. Get camera-to-chassis static TF: T_camera_base
    geometry_msgs::msg::TransformStamped tf_cam_base = tf_buffer_.lookupTransform(
        camera_frame_, robot_base_frame_, tf2::TimePointZero, tf2::durationFromSec(0.5));

    // 3. Spatial homogeneous matrix composition:
    // T_map_base = T_map_tag (YAML storage) * (T_camera_tag)^-1 * T_camera_base
    tf2::Transform T_map_tag, T_cam_tag, T_cam_base, T_tag_cam, T_map_base;
    T_map_tag.setOrigin(tf2::Vector3(map_pose.px, map_pose.py, map_pose.pz));
    T_map_tag.setRotation(tf2::Quaternion(map_pose.qx, map_pose.qy, map_pose.qz,
map_pose.qw));

    T_tag_cam = T_cam_tag.inverse();
    T_map_base = T_map_tag * T_tag_cam * T_cam_base;

    // 4. Construct and publish /initialpose initial pose topic
    geometry_msgs::msg::PoseWithCovarianceStamped pose_msg;
    pose_msg.header.stamp = stamp;
    pose_msg.header.frame_id = map_frame_;
    pose_msg.pose.pose.position.x = T_map_base.getOrigin().x();
    pose_msg.pose.pose.position.y = T_map_base.getOrigin().y();
    pose_msg.pose.pose.position.z = T_map_base.getOrigin().z();

    // 5. Dynamic covariance self-adaptation (farther distance, larger covariance)
    double distance = T_cam_tag.getOrigin().length();
    double pos_var = 0.1 + 0.05 * distance;
    double yaw_var = 0.05 + 0.02 * distance;
    pose_msg.pose.covariance[0]  = pos_var;  // x covariance
    pose_msg.pose.covariance[7]  = pos_var;  // y covariance
    pose_msg.pose.covariance[35] = yaw_var;  // yaw covariance

    initialpose_pub_->publish(pose_msg);
    return true;
}
```

5.3 Core Parameter Interpretation

Parameter Configuration File Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_nav/params/aritag_relocation.yaml
```

Core Parameter Description Table

Parameter Module	Parameter Name	Current Config Value	Core Function and Physical Meaning
AprilTag Detection	`family`	`36h11`	Tag code dictionary family (recommended to use `36h11`, highest anti-false-detection rate)
AprilTag Detection	`size`	`0.048`	Tag code black-white square physical side length (meters), used for PnP ranging calibration
AprilTag Detection	`pose_estimation_method`	`pnp`	Pose estimation algorithm (PnP perspective n-point pose solving)
AprilTag Detection	`max_hamming`	`1`	Maximum allowed Hamming error correction bits, filter incorrect recognition
AprilTag Detection	`threads`	`1`	Detection multi-threading concurrency count
Relocalization Node	`cooldown_duration`	`5.0`	Single Tag ID relocalization cooldown time (seconds), prevent high-frequency update jitter
Relocalization Node	`map_frame`	`map`	Global map reference coordinate frame
Relocalization Node	`robot_base_frame`	`base_footprint`	Robot chassis reference coordinate frame
Relocalization Node	`camera_frame`	Auto-adapted	Camera optical reference coordinate frame (`camera_color_optical_frame` / `camera`)

YAML Configuration File Complete Source

```yaml
# ============================================================================
# aritag_relocation.yaml
# AprilTag Recording and Relocalization Unified Parameter File
# ============================================================================

# ----- apriltag detection node parameters (node name: apriltag) -----
apriltag:
  ros__parameters:
    image_transport: raw        # Image transport format: "raw" or "compressed"
    family: 36h11               # Tag family name: 16h5, 25h9, 36h11
    size: 0.048                 # Default Tag side length (meters)
    profile: false              # Whether to output performance analysis information

    max_hamming: 1              # Maximum allowed Hamming distance (error correction bits)
    detector:
      threads: 1                # Thread count
```

```yaml
      decimate: 2.0             # Quadrilateral detection resolution downsampling multiplier
      blur: 0.0                 # Gaussian blur standard deviation
      refine: 1                 # Whether to refine alignment of strong gradients
      sharpening: 0.25          # Decoded image sharpening degree
      debug: 0                  # Whether to output debug images

    pose_estimation_method: "pnp"   # Tag pose estimation algorithm
    tag:
      ids:    [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20]
      frames: [tag1, tag2, tag3, tag4, tag5, tag6, tag7, tag8, tag9, tag10, tag11, tag12,
tag13, tag14, tag15, tag16, tag17, tag18, tag19, tag20]

# ----- apriltag_relocalization node parameters -----
apriltag_relocalization:
  ros__parameters:
    cooldown_duration: 5.0          # Single Tag ID relocalization cooldown time (seconds)
    map_frame: map                  # Map coordinate frame
    robot_base_frame: base_footprint  # Robot chassis coordinate frame

# ----- apriltag_recorder node parameters -----
apriltag_recorder:
  ros__parameters:
    target_frame: map               # Global coordinate frame used for recording
```